

Yang-Jun Joo, Ph.D.

Quantitative Risk Assessment Data Scientist

Autonomous Vehicle Safety | Statistical Modeling | Large-Scale Driving Data

Orlando, FL (Open to relocation to Foster City, CA) | **Email** | Phone: +1 (407) 823-2841 | **LinkedIn** | **GitHub** | **Google Scholar**

Professional Summary: Quantitative risk assessment scientist with 8+ years of research and applied project experience, including 3+ years of post-Ph.D. experience, developing probabilistic safety metrics, uncertainty-aware models, and automated analytics pipelines for autonomous-vehicle and transportation systems. Author of 15 published or accepted papers, including 9 as first or corresponding author, and Principal Investigator on two competitively funded safety research projects totaling KRW 169.6 million. Experienced with Python, SQL, large-scale driving and telemetry data, human-behavior experiments, and cross-functional delivery for public-agency safety programs.

TECHNICAL SKILLS

Programming & Data:	Python, SQL, R, pandas, NumPy, scikit-learn, PyTorch, Git, Linux, ETL and analytics pipelines, Apache Spark/PySpark.
Risk & Statistics:	Probabilistic risk modeling, Bayesian networks, reliability analysis, surrogate safety measures, regression and treatment-effect analysis, uncertainty quantification, conformal prediction, calibration, predictive intervals, experimental design.
Safety & Mobility:	Quantitative risk assessment, safety-performance metrics, AV safety evaluation, naturalistic driving video, connected-vehicle and telemetry data, traffic-conflict analysis, forward-collision warning evaluation, human factors.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar	University of Central Florida	Mar 2024 – Present
• Led quantitative safety workstreams for FDOT-sponsored projects, defining and validating safety-performance metrics using high-resolution traffic-controller, connected-vehicle, telemetry, and naturalistic driving data. • Built reproducible Python and SQL analytics pipelines supporting network-scale safety and mobility analyses across 1,050 arterial links and four Florida freeways, automating data cleaning, metric generation, validation, and dashboard-ready reporting. • Developed uncertainty-aware evaluation methods for autonomous-driving vision-language models using conformal prediction, identifying context-specific calibration failures in safety-critical driving-scene question answering. • Collaborated with transportation agencies, faculty investigators, data engineers, and research staff to translate quantitative methods into operational safety indicators and project recommendations.		
Postdoctoral Researcher	Seoul National University	Mar 2023 – Feb 2024
• Served as Principal Investigator (PI) on a KRW 73.5 million national fellowship (Sejong Science Fellowship) examining how people evaluate autonomous-vehicle risk in near-miss and safety-critical scenarios. • Designed and analyzed a controlled human-subject experiment involving 926 participants, 265 driving events, and 1,325 video clips; applied regression and heterogeneous-treatment analyses to quantify the effects of vehicle automation, event severity, and participant characteristics on perceived risk.		
Graduate Research Assistant	Seoul National University	Mar 2018 – Feb 2023
• Developed probabilistic and reliability-based risk-assessment models for autonomous-vehicle lane changes, individual-driver crash risk, and proactive collision avoidance using Bayesian networks, surrogate safety measures, and risk-field methods. • Secured and led a KRW 96.1 million Global Ph.D. Fellowship and published first-author research in Analytic Methods in Accident Research, Accident Analysis & Prevention, and Transportation Research Record.		

SELECTED QUANTITATIVE SAFETY PROJECTS

Autonomous-Vehicle Quantitative Risk Assessment

Developed generalized, probabilistic, and reliability-based risk models for proactive AV safety assessment, including risk-field modeling, Bayesian networks, and surrogate safety measures.

Human Risk and Near-Miss Evaluation

Designed a 926-participant, video-based study covering 265 near-miss events and 1,325 clips to quantify how automation attribution and event severity influence perceived risk.

Scalable Safety Analytics

Developed automated analytics for high-resolution traffic-controller and connected-vehicle data across 1,050 arterial links and four Florida freeways, producing reproducible performance metrics and uncertainty estimates.

SELECTED PUBLICATIONS

- **Joo, Y.-J.**, Kim, E. J., Park, P. Y., & Kim, D. K. (2023). A Generalized Driving Risk Assessment Method for Autonomous Vehicles Using Field Theory. *Analytic Methods in Accident Research*, 37, 100251.
- **Joo, Y.-J.**, Kho, S. Y., Kim, D. K., & Park, H. C. (2022). A Data-Driven Bayesian Network for Probabilistic Crash Risk Assessment of Individual Driver with Traffic Violation and Crash Records. *Accident Analysis & Prevention*, 174, 106721.
- **Joo, Y.-J.**, Abdel-Aty, M., Anik, B. M. T., & Islam, Z. (2025). Evaluating the Effect of Signal Adjustment on Rear-End Conflicts Using High-Resolution Event-based Data. *IEEE Transactions on Intelligent Transportation Systems*, 26(2), 123-134.
- **Joo, Y.-J.**, Kim, Y., Park, S. Y., & Abdel-Aty, M. (2026). Context-Aware Conformal Prediction for VLM-Based Driving Scene QA. *Forty-Third International Conference on Machine Learning (ICML) Workshop EMM-QA*.
- **Joo, Y.-J.**, Hwang, J., & Abdel-Aty, M. (2026). Driverless Automated Vehicles Change How People Judge Risk in Near-misses. *IEEE Transactions on Intelligent Transportation Systems* (Under Revision).

LEADERSHIP & RECOGNITION

- **Principal Investigator (PI)** on two competitively funded national research projects totaling KRW 169.6 million.
 - Recipient of the **2023 Best Paper Award** from the Transportation Research Board (TRB) of the National Academies (Standing Committee on Urban Transportation Data and Information Systems).
 - Recipient of the **2021 Best Paper Award** from the Korean Transportation Association in America (KOTTA).

EDUCATION

Seoul National University	Ph.D. in Civil & Environmental Engineering (Transportation Engineering)	Mar 2018 – Feb 2023
Seoul National University	B.S. in Civil & Environmental Engineering	Mar 2014 – Feb 2018